

INDUSTRY-GRADE EQUIPMENT FOR ACADEMIA

Engineers are created in labs. Hence it is essential to keep the lab equipment current and close to the real world to create engineers ready for tomorrow!



(Photo courtesy: SJVPMP Polytechnic, Pune)

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I still remember the excitement and contentment when I plotted the graph of diode characteristics as my first experiment at the Electronics Devices and Circuits Lab during my engineering. Before that, I thought engineering would probably be learning and mugging up of concepts and I was done with it. But it was only labs and lab studies that made me feel happy as an engineering student. As rightly said by Friedrich Engels, "An ounce of action is worth a tonne of theory."

Labs and practical implementation

of the theory play an important role in understanding the basic concepts taught in classrooms. "Academic labs provide opportunities to the engineering and science graduates to enhance their critical, analytical and creative thinking capabilities," shares Dr L.K. Maheshwari, vice chancellor, BITS Pilani. He adds, "Labs facilitate experimentation ability of students, enabling them to convert the concepts and ideas into innovation."

Realising the fact, test and measurement (T&M) manufacturers are handling the requirements from colleges, technical institutes and research insti-

tutions very innovatively. "Innovation for better learning is the policy that we follow. The T&M industry is working closely with the academia to understand and meet their needs," reveals Mohammed Ghouse, manager, Sciencetech Technologies. Manufacturers are all geared up to create more rugged, low-cost and user-friendly equipment to meet the needs of academia. Providing detailed manuals and training to the staff, and suggesting experiments and collaboration with the industry are the other steps taken by the T&M manufacturers towards creating engineers for tomorrow!

Setting up the basic electronics lab

Technical education has multiple levels. As one goes up the ladder, more advanced and specific equipment are required. However, the basic set of equipment may be needed at any level. "The basic set of equipment that are required in primary labs include multimeters, oscilloscopes, function generators and power supplies," says T.N. Singh, CEO, Salicon Nano Technology. "These are less-feature, low-cost and low-bandwidth equipment."

Elaborating the specifications of the devices, Singh adds, "For basic oscilloscopes, features include x-position, y-position, time base, less calculations and less control. As in the experiments you view signals of not more than 3 MHz to 4 MHz, normally 25MHz to 30MHz equipment is sufficient."

"A 15V/30V DC power supply is sufficient with a current of 500 mA. For higher experimentation, you look